

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

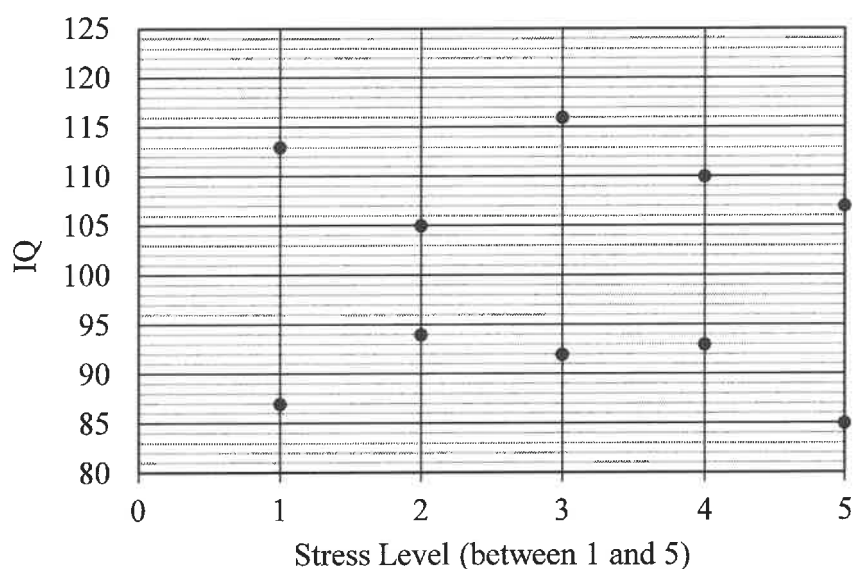
Preliminary Examination Paper 2 Answer Book

NAME: _____ MEMO

CLASS: 12 _____ TEACHER: _____

	Mark				
Question	H	T	U	Marker	HOD
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
TOTAL					
%					

QUESTION 1

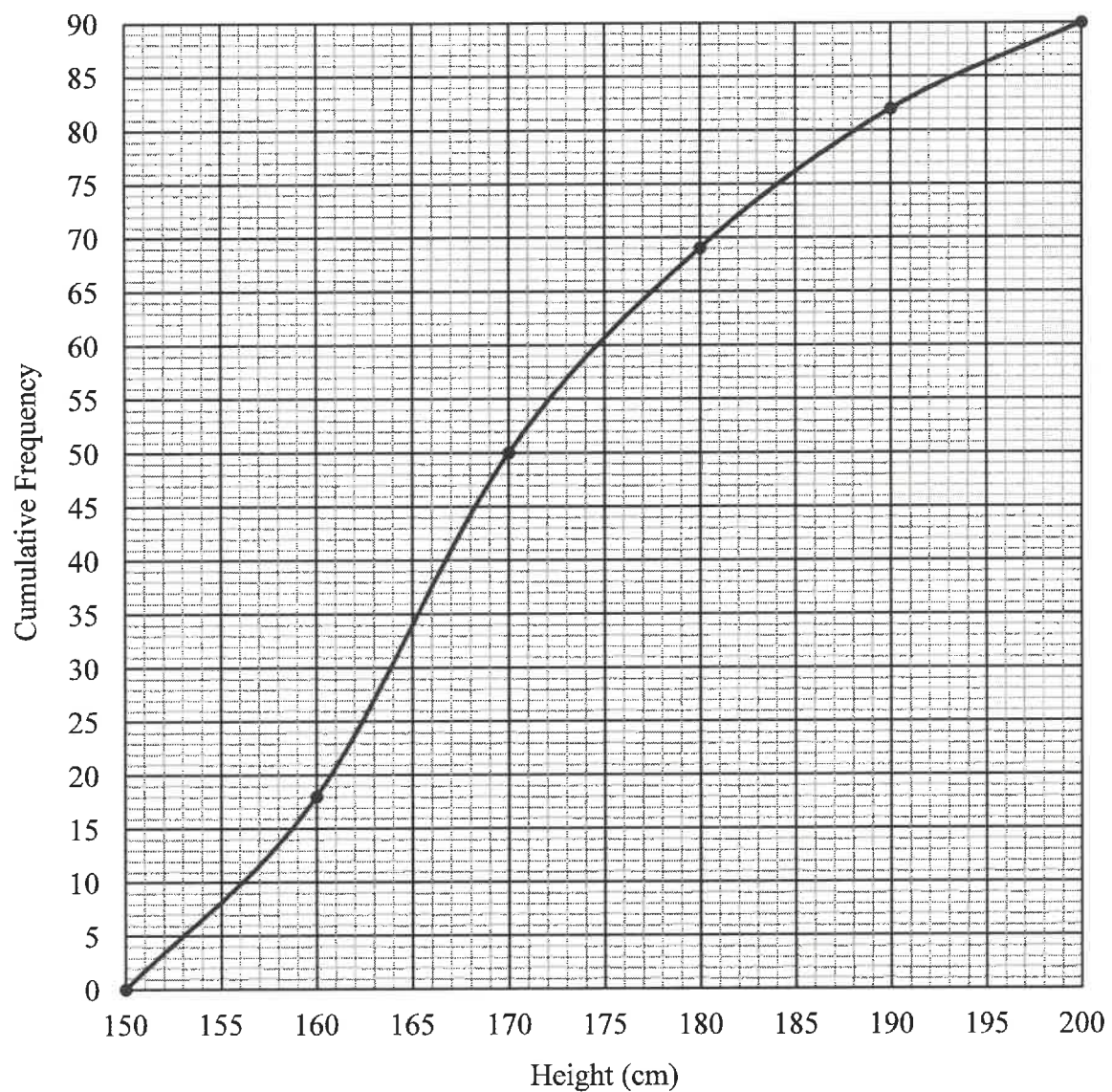


Stress Level (x)	1	3	2	5	4	4	1	2	3	5
Weekly number of hours of sleep (y)	56	45	52	33	35	36	58	50	46	37

	Solution	Marks
1.1.1	44,8 ✓✓ A	(2)
1.1.2	8,68 ✓ A	
1.2	Lower boundary: $44,8 - 8,68$ $= 36,12$ ✓CA	(2)
	∴ 3 learners ✓CA	
1.3	$\hat{y} = 62,65 - 5,95x$ ✓A ✓B	(2)
1.4	$\hat{y} = 62,65 - 5,95(4)$ ✓m $= 38,85$ hours ✓CA	(2)

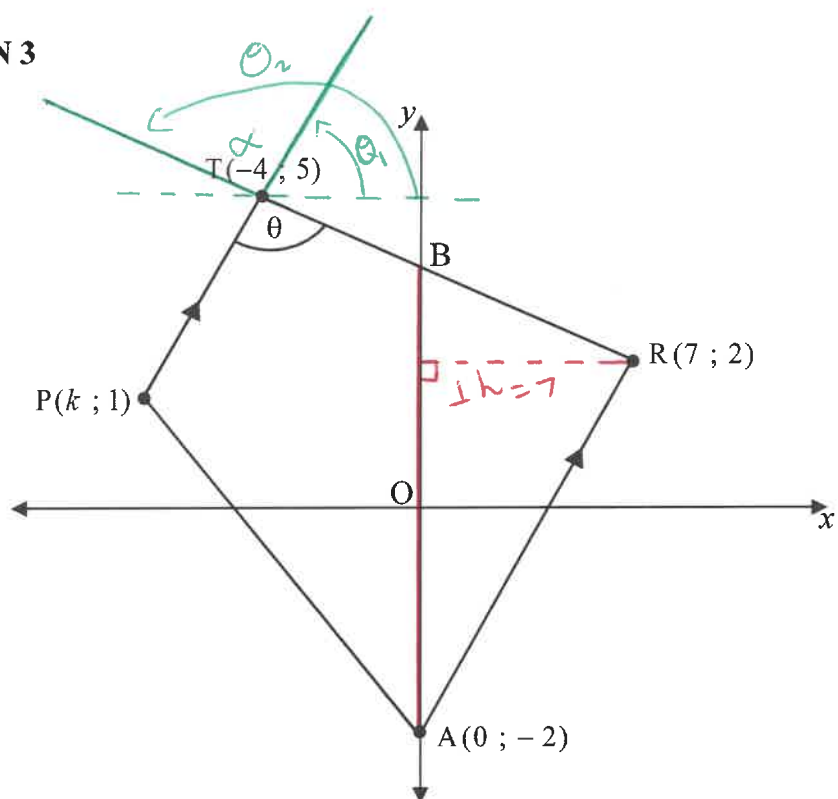
	Solution	Marks
1.5	$r = -0,97$ ✓ A	(3)
	Strong, negative, linear correlation	
	✓ ✓	
1.6	No ✓ A	(3)
	$r = -0,08$, or the scatter plot	
	shows no correlation between	
	IQ and stress levels ✓ Reason	
Any suitable motivation		[15]

QUESTION 2



	Solution	Marks
2.1	90 ✓	(1)
2.2	$160 \leq x < 170$ ✓	(1)
2.3.1	Position of $Q_2 = 45$ ✓ _m $Q_2 = 168,2$ ✓ Accept [168; 169]	(2)
2.3.2	$Q_1 = 161,5$ ✓ [161; 162] $Q_3 = 179$ ✓ [178,5; 179,5] $\therefore IQR = 179 - 161,5$ $= 17,5$ ✓ _{CA}	(3)
2.4	Number taller than 185cm = 14 ✓ _A $\therefore \% \text{ taller} = \frac{14}{90} \times 100$ ✓ _m $= 15,56\%$ ✓ _{CA}	(3)
		[10]

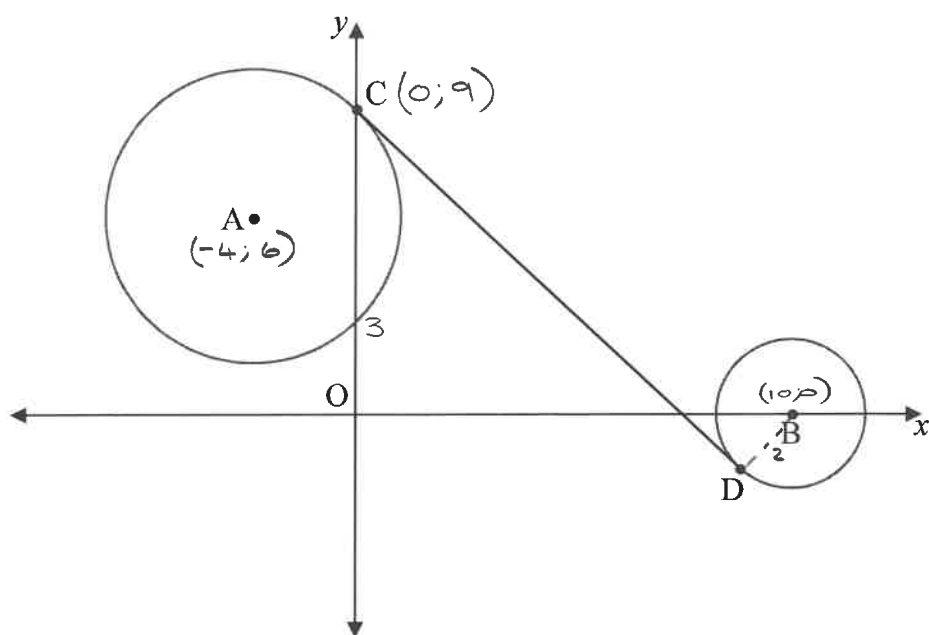
QUESTION 3



	Solution	Marks
3.1	$m_{TR} = \frac{5-2}{-4-7} = -\frac{3}{11} \checkmark m$ $y - y_1 = m(x - x_1)$ $y - 2 = -\frac{3}{11}(x - 7) \checkmark m \text{ Can also sub } (-4; 5)$ $y = -\frac{3}{11}x + \frac{43}{11} \checkmark CA$	(4)
3.2	$M(-11; 1) \checkmark A \checkmark A \text{ Diags of parm}$	(2)
3.3	$\text{Area} = \frac{1}{2}b \times h = \frac{1}{2}\left(\frac{43}{11} + 2\right)(7) \checkmark A$ $= \frac{455}{22} \approx 20,68 \text{ units}^2 \checkmark CA$	(3)

	Solution	Marks
3.4	$m_{AR} = \frac{-2-2}{0-7}$ $= \frac{4}{7} \checkmark A$ $m_{TP} = \frac{1-5}{k-(-4)}$ $= \frac{-4}{k+4} \checkmark A$ $m_{AR} = m_{TP} \quad AR \parallel TP$ $\therefore \frac{-4}{k+4} = \frac{4}{7} \checkmark m$ $-28 = 4k + 16$ $-44 = 4k$ $\therefore k = -11 \checkmark CA$	(4)
3.5	$\theta_1 = \tan^{-1}\left(\frac{4}{7}\right)$ $= 29,74^\circ \checkmark CA$ $\theta_2 = \tan^{-1}\left(-\frac{3}{11}\right) + 180^\circ$ $= 164,74^\circ \checkmark CA$ $\therefore \alpha = 164,74 - 29,74 \checkmark m$ $= 135$ $\therefore \theta = 135^\circ \checkmark \quad \text{Vert opp } \angle\text{'s}$	(4)
		[17]

QUESTION 4



	Solution	Marks
4.1.1	$x^2 + 8x + (4)^2 + y^2 - 12y + (-6)^2 = -27 + 16 + 36$ $(x+4)^2 + (y-6)^2 = 25 \checkmark A \quad \checkmark m$ $\therefore A(-4; 6) \checkmark CA \text{ both coordinates}$	(3)
4.1.2	$y^2 - 12y + 27 = 0 \quad \checkmark m \text{ let } x=0$ $(y-9)(y-3) = 0$ $y = 9 \text{ or } y = 3$ $\therefore C(0; 9) \checkmark A$	(2)
4.2	$m_{AC} = \frac{9-6}{0-(-4)} = \frac{3}{4} \checkmark CA \text{ from A and C}$ $m_{AC} \times m_{CD} = -1 \quad \text{rad} \perp \text{tan} \quad \checkmark R$ $\therefore m_{CD} = -\frac{4}{3} \checkmark CA$ $y = -\frac{4}{3}x + 9 \checkmark CA$	(4)

	Solution	Marks
4.3	$\frac{BD}{AC} = \frac{2}{5}$ $\frac{BD}{5\checkmark CA} = \frac{2}{5} \quad AC=5 \text{ from } r^2 \text{ in 4.1.1.}$ $BD = 2 \checkmark \quad \text{radius of circle B}$ $\therefore t = BD^2$ $t = 4 \checkmark CA$	(3)
4.4	$BD \perp CD \quad \frac{S}{R} \text{ rad} \perp \tan$ $\therefore BC \text{ is diameter } \frac{S}{R} \text{ chord subtends } 90^\circ$ $M \left(5 ; \frac{9}{2} \right) \checkmark CA \quad \text{Centre of } O$ $r^2 = BM^2$ $r^2 = (10-5)^2 + \left(0 - \frac{9}{2} \right)^2 \checkmark m$ $r^2 = \frac{181}{4} \checkmark CA$ $\therefore (x-5)^2 + \left(y - \frac{9}{2} \right)^2 = \frac{181}{4} \checkmark CA$	(6)
		[18]

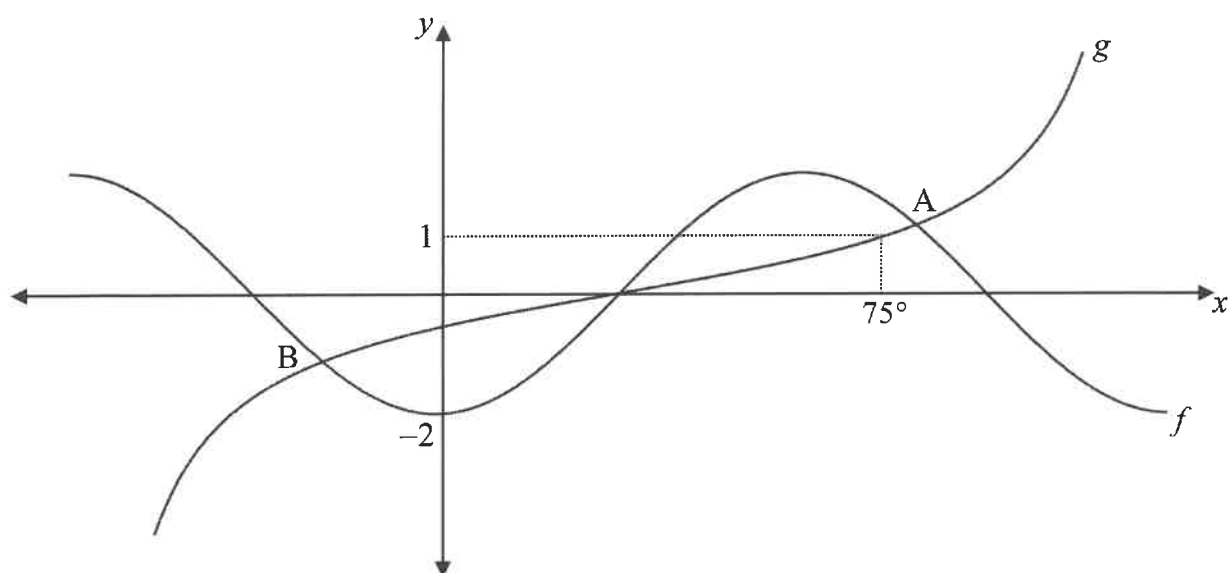
QUESTION 5

	Solution	Marks
5.1.1	$\sin x = \frac{-8}{17}$ ✓ Quad III ✓ y & r values ✓ position of $\angle y$ $x^2 = (17)^2 - (-8)^2$ $x = \pm 15$ $\therefore x = -15$ ✓A $x = -15$ $\therefore \cos y = \frac{-8}{17}$ ✓CA	(5)
5.1.2	$\tan x + \tan y$ $= \frac{-8}{-15} + \frac{-15}{-8}$ ✓CA ✓CA $= \frac{289}{120}$ ✓CA	(3)
5.2.1	$\sin 9^\circ = \sin (24 - 15)$ ✓m $= \sin 24 \sin 15 - \cos 24 \cos 15$ $= p - q$ ✓A	(2)

	Solution	Marks
5.2.2	$\cos 18^\circ = 1 - 2 \sin^2 9^\circ \quad \checkmark m \text{ dbl } \angle$ $= 1 - 2(p-q)^2 \quad \checkmark A$	(2)
5.3	$1 + \frac{\sin(90+\theta) \cdot \cos(180+\theta)}{\sin(\theta-60)\sin\theta + \cos\theta \cdot \cos(\theta-60)}$ $= 1 + \frac{\overset{\checkmark A}{\cos\theta} \cdot \overset{\checkmark A}{(-\cos\theta)}}{\cos(\theta - (\theta - 60))}$ $= 1 - \frac{\cos^2\theta}{\cos 60} \checkmark A$ $= 1 - \frac{\cos^2\theta}{\frac{1}{2}} \checkmark \text{special } \angle$ $= 1 - 2\cos^2\theta$ $= \cos 2\theta \quad \checkmark CA \text{ dbl } \angle$	(5)

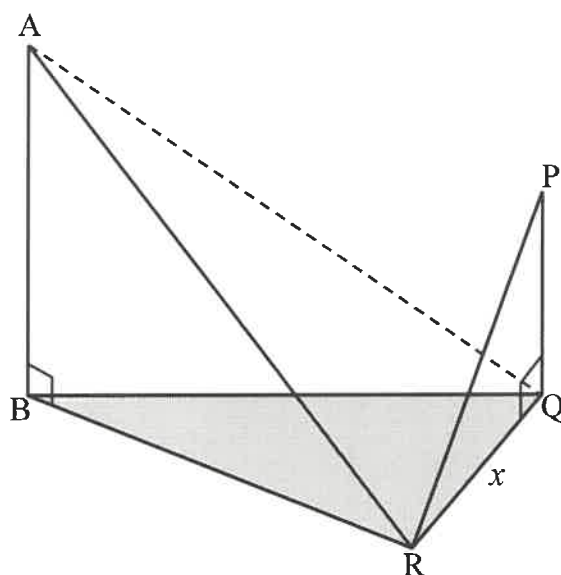
	Solution	Marks
5.4.1	$\text{LHS} = \frac{2 \tan x}{1 + \tan^2 x}$ $= \frac{2 \sin x}{\cos x} \div \left(1 + \frac{\sin^2 x}{\cos^2 x} \right) \quad \checkmark \quad \tan x = \frac{\sin x}{\cos x}$ $= \frac{2 \sin x}{\cos x} \div \left(\frac{\cos^2 x + \sin^2 x}{\cos^2 x} \right)$ $= \frac{2 \sin x}{\cos x} \times \frac{\cos^2 x}{1}$ $= 2 \sin x \cos x$ $= \sin 2x = \text{RHS}$ $\therefore \text{LHS} = \text{RHS}$	(5)
5.4.2	$x \in \{-270^\circ; -90^\circ; 90^\circ; 270^\circ\} \quad \checkmark \checkmark$	(2)
5.5	$\sin x \cos x = -0,24$ $2 \sin x \cos x = -0,48 \quad \checkmark m$ $\sin 2x = -0,48 \quad \checkmark \text{dbl } \angle$ $\text{ref } \angle = 28,69^\circ$ $\therefore 2x = 208,69^\circ + k \cdot 360^\circ$ $x = 104,35^\circ + k \cdot 180^\circ \quad ; k \in \mathbb{Z} \quad \checkmark \text{CA}$ or $2x = 331,31^\circ + k \cdot 360^\circ$ $x = 165,66^\circ + k \cdot 180^\circ \quad ; k \in \mathbb{Z} \quad \checkmark \text{CA}$ -1 penalty for $k \in \mathbb{Z}$ if left out	(4)
		[28]

QUESTION 6



6.1	$a = -2$ ✓	(3)
	$b = 3$ ✓	
	$c = 30^\circ$ ✓	
6.2	120° ✓	(1)
6.3	$x = 95$ ✓ must have $x =$	(1)
6.4	$B(-18, 5^\circ; -1, 13)$	(2)
	✓A ✓A	
6.5.1	$x = 60^\circ$ ✓	(1)
6.5.2	$-60^\circ < x \leq -30^\circ$ ✓ or	(3)
	$x = 30^\circ$ ✓ or	
	$90^\circ \leq x < 120^\circ$ ✓	
		[11]

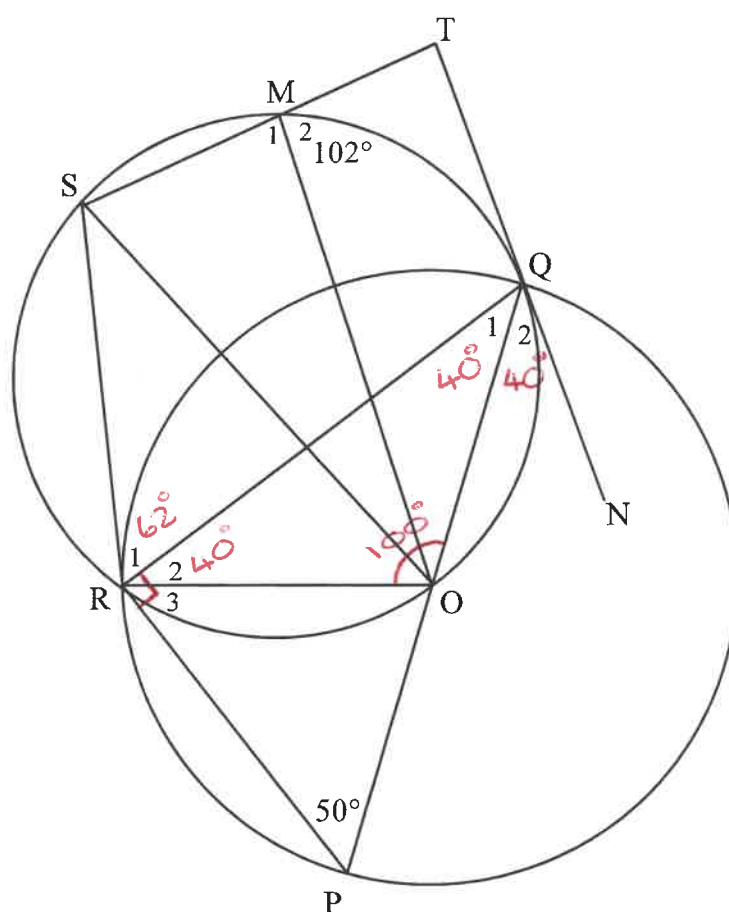
QUESTION 7



	Solution	Marks
7.1	In $\triangle QAR$:	
	$\frac{AR}{\sin(90+\theta)} = \frac{x}{\sin\theta}$ ✓sin rule	
	$\frac{AR}{\cos\theta} = \frac{x}{\sin\theta}$ ✓	
	$\therefore AR = \frac{x \cdot \cos\theta}{\sin\theta}$ ✓A	
	In $\triangle ABR$	
	$\frac{AB}{AR} = \sin 2\theta$ ✓ratio	
	$AB = AR \cdot \sin 2\theta$	
	$AB = \frac{x \cdot \cos\theta}{\sin\theta} \cdot 2 \sin\theta \cos\theta$ ✓dbl $\angle$	
	$= 2x \cos^2 \theta$	
		(5)

	Solution	Marks
7.2	In $\triangle PQR$: $\frac{PQ}{x} = \tan \theta \quad \checkmark \text{Ratio}$ $\therefore PQ = x \tan \theta$ $\therefore \frac{AB}{PQ} = \frac{2x \cos^2 \theta}{x \tan \theta} \quad \checkmark PQ$ $= \frac{2 \cos^2 \theta}{\tan \theta}$	(2)
7.3	$\frac{AB}{PQ} = \frac{2 \cos^2 30^\circ}{\tan 30^\circ} \quad \checkmark \text{subs}$ $= 2 \left(\frac{\sqrt{3}}{2} \right)^2 \div \left(\frac{1}{\sqrt{3}} \right) \quad \checkmark \text{special } \angle\text{'s}$ $= \frac{3}{2} \times \sqrt{3}$ $= \frac{3\sqrt{3}}{2} \quad \checkmark CA$	(3)
		[10]

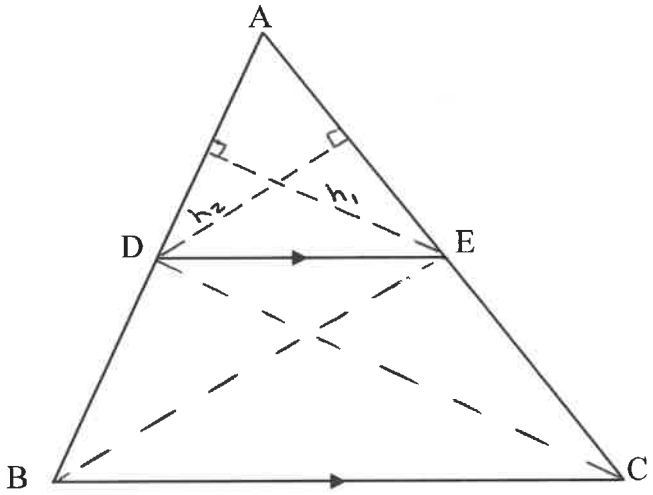
QUESTION 8



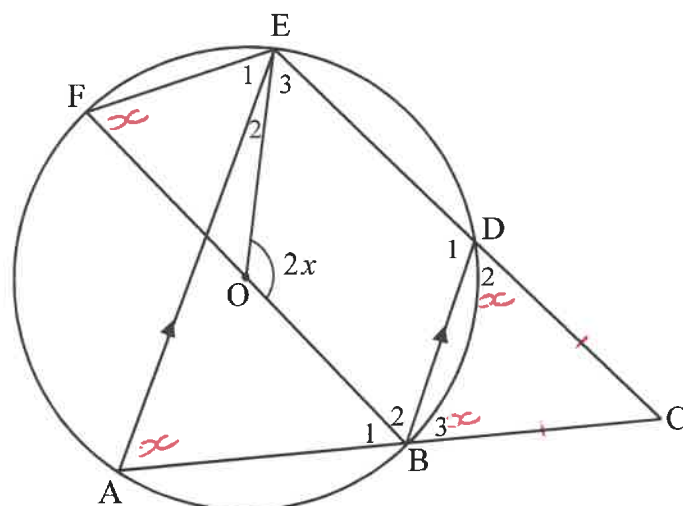
	Solution	Marks
8.1	$\hat{P}\hat{R}\hat{Q} = 90^\circ$ S/R $\angle$ in semi $\odot$	
	$\therefore \hat{Q}_1 = 40^\circ$ ✓A $\angle$'s in a Δ ✓R	
		(3)
8.2	$\hat{R}_2 = 40^\circ$ S/R $\angle$'s opp = radii	
	$\therefore \hat{R}\hat{O}\hat{Q} = 100^\circ$ S/R $\angle$'s in a Δ	
		(2)

	Solution	Marks
8.3	$\hat{Q}_2 = 40^\circ \checkmark A$ Tan chord Thrm $\checkmark R$	(2)
8.4	$\hat{R}_1 + 40 = 102^\circ$ Ext $\angle$ of cyclic quad $\checkmark R$	(2)
	$\therefore \hat{R}_1 = 62^\circ \checkmark$	
		[9]

QUESTION 9

	Solution	Marks
9.1	 Area $\triangle ADE = \frac{1}{2} AD \cdot h_1 = AD$ Area $\triangle DEB = \frac{1}{2} DB \cdot h_1 = DB$ Area $\triangle ADE = \frac{1}{2} AE \cdot h_2 = AE$ Area $\triangle DEC = \frac{1}{2} EC \cdot h_2 = EC$ but Area $\triangle DEB = \text{Area } \triangle DEC$ (same base between lines) $\therefore \frac{\text{Area } \triangle ADE}{\text{Area } \triangle DEB} = \frac{\text{Area } \triangle ADE}{\text{Area } \triangle DEC}$ $\therefore \frac{AD}{DB} = \frac{AE}{EC}$	(5)

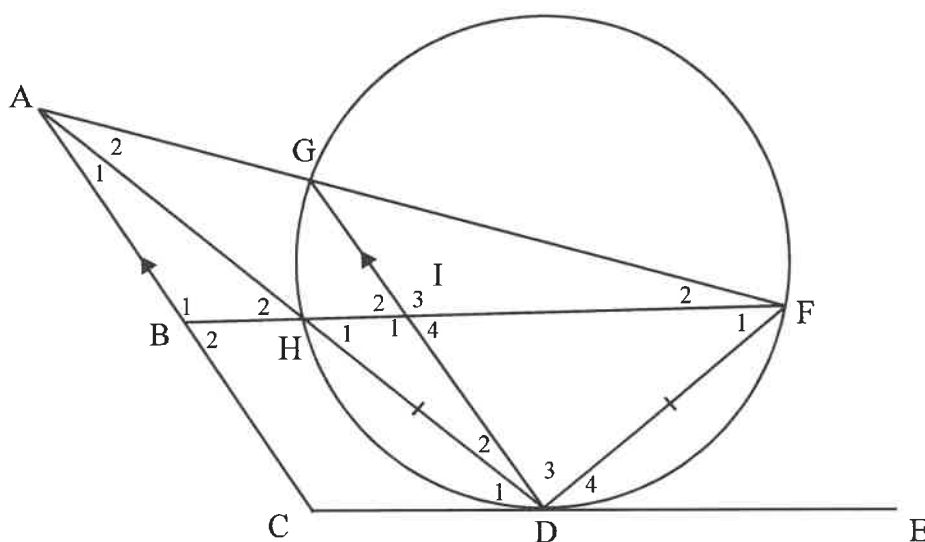
9.2



	Solution	Marks
9.2.1 (a)	$\hat{A} = x$ $\checkmark_R$ $\angle$ at centre = $2 \times \angle$ at circ	(1)
9.2.1 (b)	$\hat{F} = x$ $\checkmark_R$ $\angle$'s in same segment	
9.2.1 (c)	$\hat{D}_2 = x$ $\checkmark_R$ Ext $\angle$ of cyclic quad	(1)
9.2.1 (d)	$\hat{A}BD = 180 - x$ $\checkmark_R$ co-int $\angle$'s; $AE \parallel BD$	
9.2.2	$\hat{B}_3 = x$ $\checkmark_R$ $\angle$'s on a str line $\therefore \hat{D}_2 = \hat{B}_3$ $\therefore BC = DC$ sides opp $\checkmark_R = \angle$'s	(2)
9.2.3	$\hat{C} = 180 - 2x$ $\checkmark_R$ $\angle$'s in a Δ $\therefore \hat{EOB} + \hat{C}$ $= 2x + 180 - 2x$ $= 180 \checkmark$ $\therefore BCEO$ is a cyclic quad.	(3)
	Opp $\checkmark_R$ $\angle$'s of quad supp.	

	Solution	Marks
9.2.4	$DC = BC$ S/R line side of Δ	
	$ED \quad AB$	
	$\frac{DC}{15} = \frac{2}{3}$ ✓	
	$DC = 10\text{cm}$ ✓	
		(3)
		[17]

QUESTION 10



	Solution	Marks
10.1	$\hat{A}_1 = \hat{D}_2$ S/R Alt $\angle$'s ; $AC \parallel GD$	
	$\hat{D}_1 = \hat{F}_1$ S/R Tan chord Thrm	
	$\hat{F}_1 = \hat{H}_1$ S/R $\angle$'s opp = sides	
	$\therefore \hat{D}_1 = \hat{H}_1$	
	$\hat{C} = \hat{I}$ S/R $\angle$'s in a Δ	
	$\therefore \Delta ADC \parallel \Delta DHI$ $\angle\angle\angle$	
		(4)

	Solution	Marks
10.2	$\hat{D}_1 = \hat{H}_1$ Proven $\hat{H}_1 = \hat{H}_2$ $\checkmark R$ Vert opp $\angle$'s $\therefore \hat{H}_2 = \hat{D}_1$ $\therefore BH \parallel CD$ Corr $\angle$'s = $\checkmark R$ $\therefore \frac{AB}{AC} = \frac{AH}{AD}$ $\checkmark R$ line $\parallel$ side of Δ $\therefore AB \cdot AD = AC \cdot AH$	(3)
10.3	$\hat{A}_1 = \hat{D}_2$ Proven $\hat{D}_2 = \hat{F}_2$ $\checkmark R$ $\angle$'s in same segment $\therefore \hat{A}_1 = \hat{F}_2$ $\checkmark S$ $\therefore AB$ is a Converse Tan chord Thrm tangent	(3)
10.4	In ΔBAH and ΔBFA : $\hat{B}_1$ is common $\checkmark$ $\hat{A}_1 = \hat{F}_2$ Proven $\hat{H}_2 = \hat{A}_1 + \hat{A}_2$ 3^{rd} $\angle$ in Δ $\checkmark$ $\therefore \Delta BAH \parallel \Delta BFA$ $\checkmark$ must be in correct order $\therefore \frac{AB}{BF} = \frac{BH}{BA}$ $\checkmark R$ $\parallel \Delta$'s $\therefore AB^2 = BH \cdot BF$ $\checkmark$ $\therefore AB = \sqrt{BH \cdot BF}$	(5)
		[15]